improvement threshold that is not sufficiently large may not necessarily encourage exploration successfully. What happens when this value is set to 0.5?

6. In the previous step, we saw that setting the improvement threshold to an appropriate value is crucial for PoI. However, it's not obvious how to do this across multiple applications and objective functions. A reasonable heuristic is to dynamically set it to some percentage of the incumbent value, specifying that we would like to see *1 +* increase in the incumbent value. Implement this in our code with a 110 percent improvement requirement.

In the next section, we will discuss the second approach to addressing to overly exploitative behavior of PoI, which is to take into account not only whether a candidate data point will yield an improvement, but also *how much* we expect the improvement to be.

4.3 Optimizing the expected value of improvement

As we saw in the previous section, naïvely seeking to improve from the incumbent leads to over-exploitation from PoI. This is because by simply moving away from the incumbent by a small amount in an appropriate direction, we can achieve a high probability of improvement. Therefore, optimizing this probability of improvement is *not* what we want to do. In this section, we will learn to further account for the *magnitude* of the possible improvements that we may observe. In other words, we also care about how much improvement we can make from the incumbent. This will give us one of the most popular BayesOpt policies called *Expected Improvement* (EI).

The motivation for seeking to account for the magnitude of the improvement is clear. Consider the example in Figure 4.15.

The left panel shows the computation made by the PoI policy, which considers only whether each candidate data point improvements from the incumbent. Therefore, a point that improves by a small amount and one that significantly improve from the incumbent are treated equally.

On the other hand, the right panel shows what happens if we were to also consider the magnitude of the possible improvements. Here, although points x_1 and x_2 are still treated as bad (since they don't improve from the incumbent x_0), x_4 is considered better than x_3, as the former offers a larger improvement. Similarly, x_5 is considered the best out of the five candidate points.

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